

# Functional & Performance Testing Report

## GenAI / Automation Model Test Scenarios & Execution Results

|                |                                                |
|----------------|------------------------------------------------|
| Date:          | 04 October 2026                                |
| Team ID:       | SWTID-2026-5601                                |
| Project Name:  | Auto Ticket Classification using Flow Designer |
| Maximum Marks: | 4 Marks                                        |

### 1. Test Plan & Scope Overview

This document summarizes the Functional Testing (FT) and Performance Testing (PT) scenarios executed for the Auto Ticket Classification System utilizing ServiceNow Flow Designer and Natural Language Understanding (NLU). Testing validates input handling, automatic classification logic, REST API connections, response latency, and system behavior under load.

### 2. Test Scenarios & Results

| Test Case ID | Scenario (What to test)                                                           | Test Steps (How to test)                                                                                                 | Expected Result                                                                      | Actual Result                                                                       | Pass/Fail |
|--------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------|
| FT-01        | Text Input Validation (e.g., ticket description, issue summary)                   | Enter valid and invalid text in ticket description input fields (special characters, empty text, script tags)            | Valid inputs accepted; invalid inputs produce clear user-friendly error messages     | Form validated input correctly and sanitized special characters                     | Pass      |
| FT-02        | Number Input Validation (e.g., priority scale, urgency rank, ticket count limit)  | Enter numbers within (1-4) and outside (0, 99) the valid priority/urgency range                                          | Accepts valid priority values (1-4); shows error for out-of-range inputs             | Input restricted to 1-4 scale; out-of-range values rejected gracefully              | Pass      |
| FT-03        | Content Generation / Auto-Classification (e.g., Category, Subcategory prediction) | Provide complete ticket descriptions (e.g., 'Wi-Fi keeps dropping in Room 302') and trigger classification               | Correct Category ('Network') and Subcategory ('Wireless') generated based on input   | Flow Designer assigned Category: Network, Subcategory: Wireless with 94% confidence | Pass      |
| FT-04        | API Connection Check                                                              | Verify REST API connection between Service Portal, Flow Designer, and NLU Engine using authenticated API key/OAuth token | API responds successfully with HTTP 200 OK and valid JSON response                   | API connected successfully; authenticated token validated                           | Pass      |
| PT-01        | Response Time Test                                                                | Use automated performance timer to measure end-to-end ticket classification and routing time                             | Classification and ticket creation process completes under 3 seconds                 | Average processing and classification time recorded at 1.42 seconds                 | Pass      |
| PT-02        | API Speed Test (Concurrent Requests)                                              | Send 50 concurrent API classification requests simultaneously using JMeter / API test suite                              | API processes concurrent calls without throughput degradation or connection timeouts | Handled 50 concurrent requests with 0% error rate and consistent throughput         | Pass      |

| Test Case ID | Scenario (What to test) | Test Steps (How to test) | Expected Result | Actual Result | Pass/Fail |
|--------------|-------------------------|--------------------------|-----------------|---------------|-----------|
| PT-03        |                         |                          |                 |               |           |